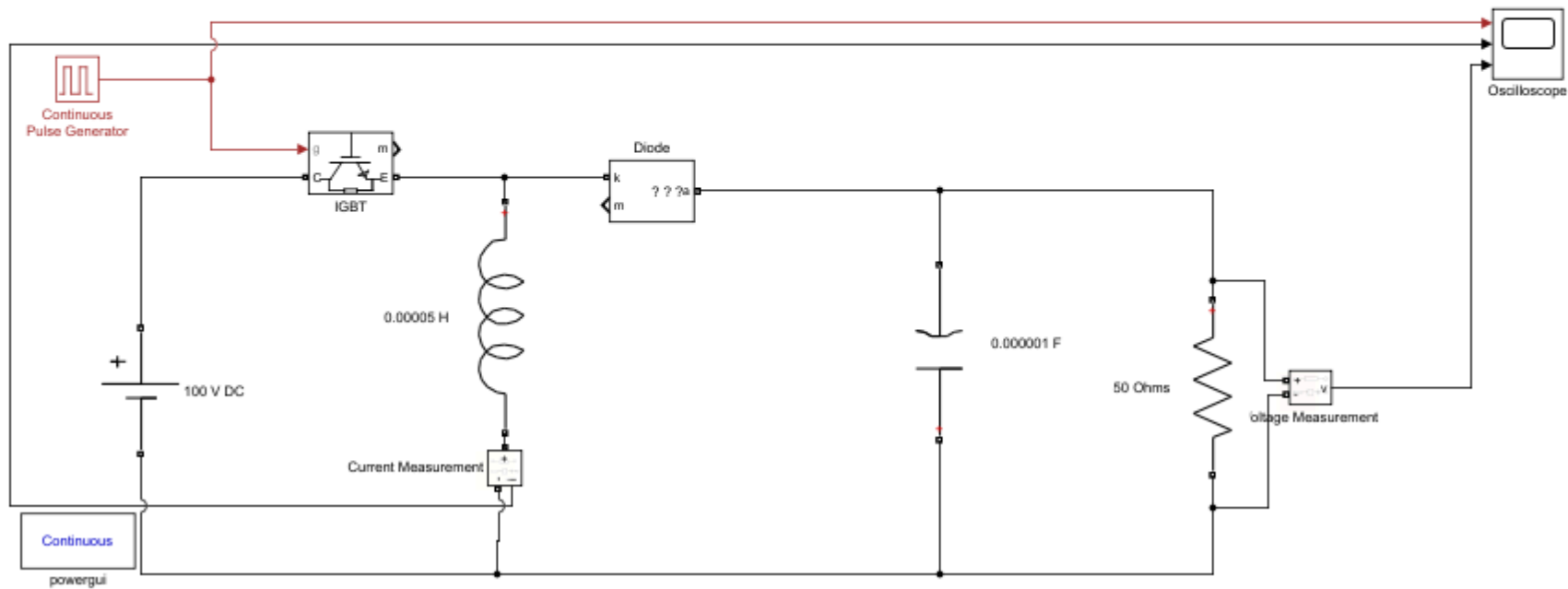


Buck_Boost_Converter

Details for Buck_Boost_Converter

Buck Boost Converter

By,
Gowravam Sandeep Kumar
Dept of EEE
RGUKT Srikakulam



SANDEEP GOWRAVAM

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Model - Buck_Boost_Converter

Full Model Hierarchy

1. [Buck Boost Converter](#)

Simulation Parameter	Value
Solver	VariableStepAuto
RelTol	1e-3
Refine	1
MaxOrder	5
ZeroCross	on

Table 1. Sample Times

Color	Annotation	Description	Value
	Cont	Continuous	0
	FiM	Fixed in Minor Step	[0,1]
	V1	Variable 1	Continuous Pulse Generator
	Inf	Constant	Inf

[\[more info\]](#)

System - Buck_Boost_Converter

Buck Boost Converter

By,
Gowravam Sandeep Kumar
Dept of EEE
RGUKT Srikakulam

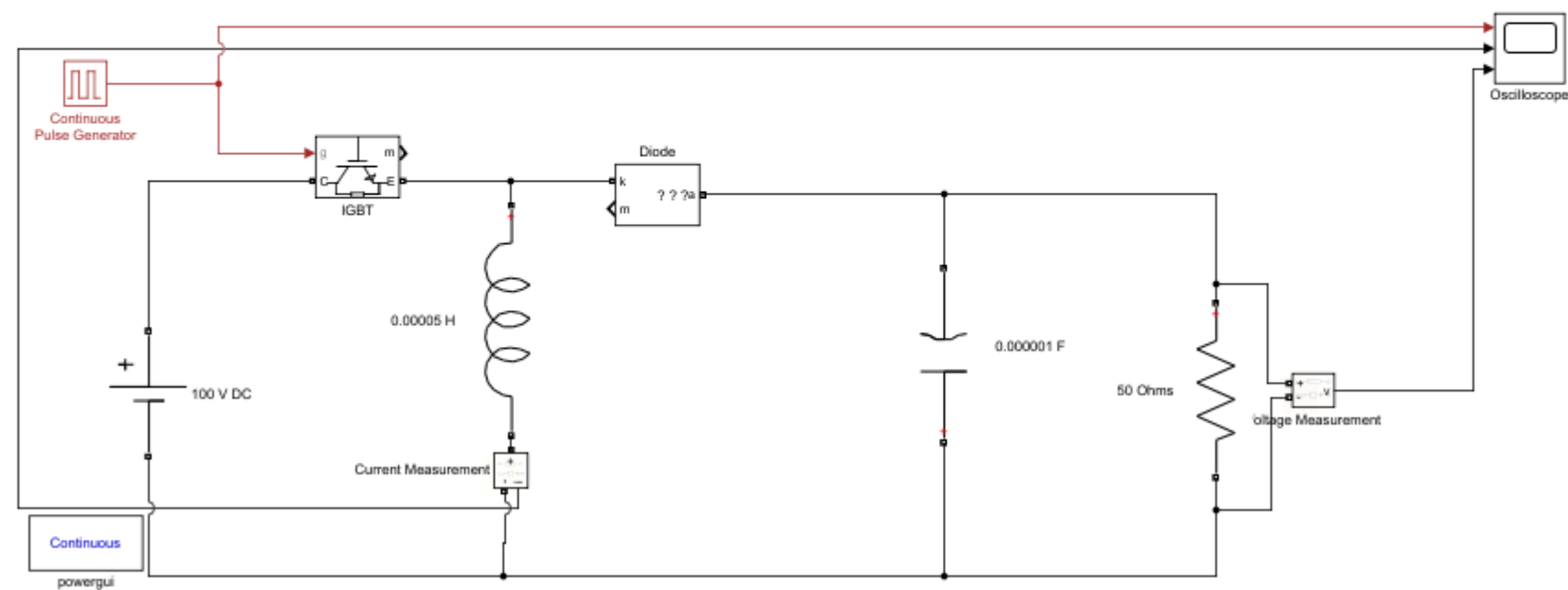


Table 2. Current Measurement Block Properties

Name	Output Type
Current Measurement	Complex

Table 3. DC Voltage Source Block Properties

Name	Amplitude	Measurements
100 V DC	100	None

Table 4. Diode Block Properties

Name	Ron	Lon	Vf	IC	Rs	Cs	Measurements
Diode	0.001	0	0.8	0	500	250e-9	on

Table 5. DiscretePulseGenerator Block Properties

Name	Pulse Type	Time Source	Amplitude	Period	Pulse Width	Phase Delay	Sample Time
Continuous Pulse Generator	Time based	Use simulation time	2	1/25000	50	0	1

Table 6. IGBT Block Properties

Name	Ron	Lon	Vf	IC	Rs	Cs	Measurements
IGBT	0.001	0	1	0	1e5	inf	on

Table 7. PSB option menu block Block Properties

Name	Simulation Mode	Iterations	Frequencyindice	Pbase	Err Max	Units V	Units W	Function Messages	Echomessages	Enable Use Of TLC	X0status
powergui	Continuous	50	0	100e6	1e-4	kV	MW	off	off	off	blocks

Table 8. Series RLC Branch Block Properties

Name	Branch Type	Capacitance	Setx 0	Measurements
0.000001 F	C	0.000001	off	None
0.00005 H	L	1e-6	off	None
50 Ohms	R	1e-6	off	None

Table 9. Voltage Measurement Block Properties

Name	Output Type
Voltage Measurement	Complex

Appendix

Table 10. Block Type Count

BlockType	Count	Block Names
Series RLC Branch (m)	3	0.000001 F , 0.00005 H , 50 Ohms
Voltage Measurement (m)	1	Voltage Measurement
Scope	1	Oscilloscope
PSB option menu block (m)	1	powergui
IGBT (m)	1	IGBT
DiscretePulseGenerator	1	Continuous Pulse Generator
Diode (m)	1	Diode
DC Voltage Source (m)	1	100 V DC
Current Measurement (m)	1	Current Measurement

